



Normative isometric plantarflexion strength values for professional level, male rugby union athletes

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ABSTRACT

Objectives: The primary aim was to establish normative values of isometric plantarflexor muscle strength in professional male rugby union players and compare forwards with backs. The secondary aims were to examine how individual playing position or age influences isometric plantarflexor strength.

Design: Cross-sectional.

Setting: Testing at professional rugby clubs.

Participants: 355 players (201 forwards and 154 backs) from 9 clubs in the English Premiership club competition.

Main outcome measures: Maximal unilateral isometric plantarflexion strength was measured, using a Fysiometer C-Station, in a seated position with a flexed knee and in maximal available dorsiflexion. Values are reported normalised to body mass and specific to playing position.

Results: Mean combined limb isometric plantarflexion strength for the group was 193.1 kg (SD 32) or 1.86 xBW. (SD 0.31). Forwards were significantly weaker than backs (forwards = 1.75xBW (SD 0.26), backs = 2.00xBW (SD 0.28) ($p < 0.0001$)). Age category revealed no influence on plantarflexor strength.

Conclusion: This study presents normative isometric plantarflexion strength values for professional male rugby union players. Forwards are typically relatively weaker than backs.

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1. Introduction

The triceps surae is a relatively complex muscle group with a large physiological cross-sectional area (Albracht et al., 2008), the soleus in particular comprises 60% of the triceps surae and produces the majority of the plantarflexor forces needed for locomotion (Dorn et al., 2012). Running requires plantarflexor muscles to operate at a high proportion of their maximal capacity across a range of speeds (Kulmala et al., 2016). This demand potentially

places these tissues at an elevated risk of fatigue related injury in athletes participating in sports characterised by high intensity or high volume running (Willer et al., 2021). The calf muscles, and soleus in particular, are known to be the greatest contributors to acceleration during running (Hamner & Delp, 2013; Pandey et al., 2021) with peak forces reaching up to 16 times body mass (16xBW) (Dorn et al., 2012). During high-speed running the plantarflexors are theorised to function relatively isometrically (Kawakami et al., 2002). This allows the tendon to contribute to energy recoil, effectively increasing force production.

The plantarflexor complex has also been found to be highly influential in direction change activities such as cutting and pivoting (Maniar et al., 2022). Biomechanical analysis of cutting

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performance has shown peak plantarflexion power to be the strongest determinant of cutting time (Marshall et al., 2014) and higher levels of plantarflexion force and Achilles tendon stiffness are related to faster turn times (Houghton et al., 2013).

Injuries affecting the calf muscles and Achilles tendon complex are a significant problem for those participating in running based, multidirectional team sports (Green et al., 2020a; Hägglund et al., 2013). Rugby union is an intermittent, high-intensity contact sport during which players typically cover up to 7 km during a match (Bridgeman & Gill, 2021). Calf muscle injuries have been one of the two most common and one of the two highest burden training injuries in Premiership rugby in recent seasons (Kemp et al., 2019). Typical time loss has been 51 days (29–84 days) absence (Kemp et al., 2017). These injuries are experienced by players in all playing positions, however it is thought that they disproportionately affect hookers, risk ratio of 1.5 compared to other forwards and tighthead props, risk ratio of 5.85 compared with other forwards (Brooks & Kemp, 2011). Centres were also identified as at greater risk than other backs, risk ratio of 0.98 (Brooks & Kemp, 2011). Forwards are reported as experiencing 5.7 calf injuries per 1000 player hour whilst backs experience only 3 per 1000 h (Kemp et al., 2018) with calf injuries causing 77 days time-loss per 1000 h (Brooks et al., 2005).

Injuries involving the soleus muscle have been found to be significantly more common than those of the gastrocnemius muscle in sports characterised by high volumes of multidirectional running such as Australian Football, where soleus injuries account for 84.6% of all calf muscle injuries (Green et al., 2020a; Waterworth et al., 2017). In contrast, more intermittent sports characterised by lower running volumes such as American Football (Wellman et al., 2016) have found gastrocnemius injuries to be more prevalent (Werner et al., 2017). Given the contribution of soleus to running activities, this contrast between sports of differing characteristics is unremarkable. However, it serves to emphasise the relevance of the demands of specific sports to resultant injury risks. Soleus muscle injuries are characterised by high recurrence rates (Green et al., 2020a) with these reinjuries being shown to affect older players significantly more than younger players and with the soleus accounting for 32/35 reinjuries (Green et al., 2020a, 2020b). Older footballers have also been shown to be at greater risk of first calf injuries than younger players to a larger degree than other lower limb muscle injury (Ekstrand et al., 2011).

Young players (16–21 years) experience 0.32 calf muscle injuries per 1000 h, players in aged 22–30 experience 1.07 and players over 30 years old experience 1.89 per 1000 h (Ekstrand et al., 2011).

Plantarflexor weakness is well associated with injuries to the calf complex (Green & Pizzari, 2017) and was believed to be a key intrinsic risk factor for calf muscle strain injuries by a group of expert sport clinicians working with elite (professional) team and individual sport athletes (Green et al., 2022). However, as is the case with other lower limb muscle injuries, evidence for strength as a risk factor remains inconclusive (Freckleton et al., 2014) and has not been investigated appropriately.

Our previous research on participants with Achilles tendinopathy has identified that the plantarflexor strength on the uninjured (asymptomatic) limb is no different from the injured (symptomatic) limb. However, when compared to healthy controls it is substantially lower (O'Neill et al., 2019). Therefore using the other limb as a target for rehabilitation may not be useful for problems relating to the plantarflexors. We have found the identical relationship in our extensive but unpublished calf injury and screening work, highlighting the requirement for healthy normative values.

Normative isometric strength values are particularly valuable for sports medicine practitioners in guiding return to play decisions

and allowing for greater detail in individualised performance programming and injury risk mitigation input. Given the significant variance in positional demands of Rugby Union (Bridgeman & Gill, 2021; Lindsay et al., 2015) it is essential that the normative strength scores are specific not only to a professional rugby union population but also to the individual playing positions concerned.

1.1. Aims

The primary aim was to establish normative isometric plantarflexion strength for professional rugby union players relative to their body mass and to compare these between positional categories (Forwards and Backs).

The secondary aims were to both examine the normalised isometric plantarflexion strength scores by individual playing position and also to compare these scores by age group category.

2. Methods

This study utilised a cross-sectional design and adhered to the STROBE reporting standards (Strengthening the Reporting of Observational studies in Epidemiology) (Cuschieri, 2019) and the CHAMP statement (Checklist for statistical Assessment of Medical Papers) (Mansournia et al., 2021).

2.1. Participants and ethics

This study collected isometric plantarflexion strength data from 355 players. All players were male, full-time professional rugby union players in the English Premiership at time of testing and were recruited from nine Premiership clubs. All players at these clubs who were deemed fit to complete testing by club medical staff, and whom subsequently consented to testing were eligible for participation. All participants were given a 'Participant Information Sheet' before consenting to being involved in testing. Ethical approval was granted by the College of Life Science Committee for Research Ethics Concerning Human Subjects at the University of Leicester.

2.2. Public and patient involvement

Key stakeholders (players, medical staff and conditioning staff) at a number of the involved clubs along with players from one of the clubs were consulted throughout the design process for the project.

2.3. Testing

The testing position, equipment and protocol replicated a previously validated process (O'Neill et al., 2023). This procedure used the Fysiometer C-Station (FysioMeter, Denmark) system which has a test re-test reliability ICC score of 0.91 (95%CI 0.86–0.94) and good Pearson correlation with isokinetic dynamometry (r value = 0.72 : 95%CI 0.52–0.84). Fig. 1 shows the setup position for the isometric test using the C-Station. The FysioMeter® software (Broenderslev, Denmark) analyses data from all transducers at 100 Hz simultaneously, with a 4th order Butterworth filter (cut-off 20 Hz). The device has an accuracy of 0.98N whilst measuring from 0 to 2943N.

All data were collected at each club during the first two days of a training week. With the majority (277 subjects) of data being collected in the late weeks of pre-competition phase (pre-season) and remainder being collected in the early weeks of the competition phase (in-season). The characteristics of training in these two phases did not typically differ greatly and thus this small change in



Fig. 1. Testing position using Fysiometer C-station for isometric plantarflexion strength test.

exposure is unlikely to have affected the strength scores.

2.4. Testing set-up

All players were tested in a seated position, with the knee at 90–110° flexion and ankle in full available dorsiflexion. Players were tested in normal gym training footwear with the metatarsal heads of the tested limb positioned centrally on the forceplate to the marked medial-lateral (ML) line with the foot centred to an anterior-posterior (AP) line. The 2nd/3rd metatarsal heads were centred over the ML and AP line transect point. Once in position, a seatbelt running from under the force plate platform was applied directly over the femoral epicondyles, avoiding compression of the quadriceps tendon, and this was tightened to ensure a secure setup with no change of position. Pre-tension was standardised to 30 kg. Pre-tensioning of the belt is necessary to “lock” the participant into maximum available dorsiflexion, individualised to the player. The warm-up sub-maximal movements normally cause a small increase in tissue lengthening and relaxation reducing the pre-tension reading to the weight of the limb-around 15 kg. The system is not re-calibrated after the pre-tension and this value is included in the overall force measure. Participants did not require cushioning between the belt and knee in this set-up. All testing was completed by the same member of the research team who was trained and familiar with the equipment. All testing was completed with the ankle in maximal available dorsiflexion for each individual as this position produces maximal force (Baxter & Piazza, 2014; Ferris et al., 2020).

2.5. Testing procedures

Once positioned and secured, players completed three sub-maximal, isometric contractions serving to act as a familiarisation period. Players then completed three maximal isometric contractions of 3–4 s with a rest interval of 3–5 s between efforts and the belt being re-tensioned if necessary. Players were then repositioned to test the contralateral limb with the same testing procedure being followed. The left side was tested first for all subjects. This order was chosen as previous extensive pilot testing showed no order effect of testing and the act of standardising the test order allowed for improved data collection speed and reduces any recording/inputting errors related to limb tested.

Isometric testing was used as it is reliable and valid but also very quick to complete and portable. For many years the Isokinetic dynamometer has been considered the “gold standard” however it requires expensive equipment, is unportable and would make this study (size and geographic spread) impossible. The Calf Raise test could also be used and would be appropriate, but would require endurance testing and would induce a degree of delayed onset muscle soreness (Hébert-Losier et al., 2009). This was not acceptable to the involved rugby clubs and players as it would impede training. Retrospective analysis of previously published data (O'Neill et al., 2019) shows a strong correlation between concentric and eccentric isokinetic and isometric test results (Pearson correlation of >0.6) and previous work has demonstrated strong relationships between isometric plantarflexion force and physical performance such as sprinting (Houghton et al., 2013).

2.6. Analyses

All test data was recorded in the native Fysiometer software and exported into Excel. The peak reading from the three maximal isometric contractions for each limb was used for analyses. The analysis was completed on the peak absolute forces reported in KG rather than Newtons to allow easier comparison for individuals body mass. Force data was then calculated as a representation compared to their body mass - i.e. force (Kg)/body mass (Kg), presented as xBW. The xBW metric is more appropriate for weight bearing muscles functioning during locomotion such as the plantarflexors. The plantarflexors will produce significant internal forces during locomotion and this is relevant to body mass not absolute force. The xBW metric allows easier and quicker comparison across playing cohorts where body mass varies extensively.

The metric is also more suitable for clinical interpretation as it creates a simple measure i.e. 2xBW would equate to a force double that of the person's body mass.

The Minimal Detectable Change of the Fysiometer C-station is 14.9 kg in a mixed gender population and will be used to determine if there is an actual difference between groups in the analysis (O'Neill et al., 2023).

Data were analysed using R (version 4.1.3). The primary outcome of the study was to compare normalised combined limb strength between positional categories (Forwards and Backs). Secondary outcomes were to examine normalised limb strength by specific playing position and by age group. Age was recorded as a continuous variable but was later categorised to allow for comparison between age groups. This was completed to allow comparison with previously published data showing higher risk of muscle injury with older athletes (Ekstrand et al., 2011).

Demographic data were investigated by summarising characteristics. For the primary outcome, a two-sample independent *t*-test was used to test for a difference in calf strength between positional category. Descriptive statistics were used to examine calf strength by playing position. A one-way ANOVA was performed to

test for differences in calf strength by age group. Post-hoc analysis was performed, using Tukey's method to adjust for multiple testing, to investigate pairwise comparisons by age group. Results were statistically significant if $p < 0.05$.

The distribution of the data by positional category, specific position and by age group was assessed for normality using visual inspection of histograms. Boxplots were produced in SPSS (version 26) to show the spread of the population.

3. Results

3.1. Primary outcome - normative strength by positional category (forwards and backs)

All playing positions were included at all clubs (forwards = 201, backs = 154). Participant demographics for each positional group are shown in Table 1.

In total 355 subjects were tested, the overall group means were age = 24 years (SD 4.6) and body mass = 105 kg (SD 14.2).

Overall group mean plantarflexor force = 193.1 kg (SD 32) for 703 limbs in the 355 subjects. In the case of seven participants, testing was only completed on a single limb due to recent injury to the other leg. These players were all considered fully fit for training and matchplay. These seven participants are not included in combined limb strength data results ($N = 348$) but are included in the single limb data ($N = 355$).

Mean Right isometric plantarflexion force for whole group was 195 kg (SD 32.1 kg) and Left was 191.2 kg (SD 31.9 kg). When normalised to body mass, mean Right was 1.88 xBW (SD 0.31BW) and mean Left was 1.84 x BW (SD 0.31 x BW).

Comparison of normalised combined limb strength by positional category is shown in Table 2. Forwards were found to be significantly weaker than backs. The difference between forwards and backs of 15.6 kgs also exceeds the identified MDC for the device, MDC = 14.9 kgs (O'Neill et al., 2023), suggesting both statistical and meaningful differences between the positions. The difference suggests forwards are stronger when measuring force alone, however it is important to correct the strength measure for BW and when this has been done the Backs are stronger.

When positions were examined according to positional category (forwards and backs), data showed that the normalised (BW) combined limb strength of forwards (1.75 x BW (SD 0.26 x BW)) was relatively weaker than that of backs (2.00 x BW (SD 0.28 x BW)). Independent Samples T-Test showed this to be significantly different with a p value < 0.0001 . The distribution of this data is shown in Fig. 3 and shows relatively little overlap between the groups.

3.2. Secondary outcome 1 - normative calf strength by individual playing position

Descriptive statistics are presented in Table 3 reporting the absolute and normalised strength values by playing position. The distribution of this data is shown in boxplots (Fig. 2) to allow visualisation of populations and easier comparison.

3.3. Secondary outcome 2 – difference in normative calf strength by age group

The summary of normalised combined limb strength by age category is presented in Table 4. The differences between these age categories are presented in Table 5.

4. Discussion

This is the first study assessing the isometric plantarflexion strength of professional, male rugby union players. The primary finding of the study is that when data is normalised to individual body mass, forwards are significantly weaker than backs. It is critical to locomotive muscles to normalise to body mass as these muscles are responsible for moving players/athletes around and heavier players require more force to move.

Broadly, the sampled population demonstrated mean isometric plantarflexion strength of 1.86xBW (SD 0.29). Testing beyond 80° knee flexion position primarily assesses the capacity of soleus as the potential for gastrocnemius to contribute is inhibited (Laufer et al., 2014). A strength measurement taken in this position has clear practical relevance to this cohort as the role of soleus is well demonstrated in contributing to dynamic multidirectional activities, such as during rugby training and play, whilst also having clinical relevance to both calf muscle injuries (Green et al., 2020a) and Achilles tendinopathy (O'Neill et al., 2019). Rugby and other sports injuries data also highlights that the soleus is the most injured of the calf muscles (Green et al., 2020a; Kemp et al., 2019). This specific relevance makes the measurement valuable in both identifying potentially 'at risk' individuals and also those for whom targeted performance development strategies are required.

The running and contact demands of rugby union varies between players in different playing positions (Bridgeman & Gill, 2021). Backs are known to cover greater distance (both absolute and relative) and also greater distance at high-speed than forwards (Bridgeman & Gill, 2021; Lindsay et al., 2015). Forwards are generally involved in more contact events (impacts and collisions) than backs (Bridgeman & Gill, 2021; Lindsay et al., 2015). Within both positional groups, more subtle differences also exist between individual positions; for example back row forwards cover

Table 1

Demographic data for individual position, as well as positional category. Numbers given as number (%) and age and weight given as mean (standard deviation).

Position	Number of observations (left limb)	Number of observations (right limb)	Number with ≥ 1 measurement	Age (years)	Body mass (kg)
Forwards	200 (56)	196 (55)	201 (57)	24.2 (4.6)	114.38 (9.77)
Loosehead Prop	31 (9)	31 (9)	31 (9)	24.3 (4.4)	117.3 (9.76)
Hooker	29 (8)	29 (8)	29 (8)	23.6 (4.6)	105.91 (6.69)
Tighthead prop	32 (9)	32 (9)	32 (9)	24.8 (4.5)	122.67 (6.41)
Lock	42 (12)	39 (11)	42 (12)	24.1 (5.2)	118.3 (7.35)
Flanker	45 (13)	45 (13)	45 (13)	23.8 (4.5)	107.37 (6.52)
No8.	21 (6)	20 (6)	22 (6)	25.1 (3.7)	116.19 (10.1)
Backs	153 (43)	154 (43)	154 (43)	24.2 (4.7)	92.78 (8.68)
Scrum Half	31 (9)	31 (9)	31 (9)	25.3 (4.8)	84.96 (6.43)
Fly Half	20 (6)	20 (6)	20 (6)	23.4 (4.4)	88.31 (4.77)
Centre	41 (12)	41 (12)	41 (12)	24.2 (4.8)	98.2 (6.56)
Wing	38 (11)	39 (11)	39 (11)	23.1 (4.5)	96.03 (9.79)
Full back	23 (6)	23 (6)	23 (6)	24.8 (5)	92.03 (5.05)

Table 2
Comparison of normalised combined limb strength by positional category (i.e. Forwards and Backs). P-value is a result of the two-sample independent *t*-test performed.

Positional Category	Number	Mean normalised combined strength	Difference in means	95% Confidence Interval	P-value
Forwards	195	1.75	−0.243	(−0.300, −0.185)	<0.0001
Backs	153	2.00			

Table 3
Descriptive statistics of the average force and normalised left, right and combined limb strength for each playing position. Values are given as mean (standard deviation).

Playing Position	Number of obs. with both limb measurements	Force (kg)	Normalised combined strength (x BW)	Left limb		Right limb	
				Force (kg)	Normalised combined strength	Force (kg)	Normalised combined strength
Forwards	195	199.8 (31.3)	1.75 (0.26)	197.8 (33.4)	1.7 (0.28)	202.14 (32.6)	1.77 (0.27)
Loosehead Prop	31	207.8 (32.9)	1.78 (0.27)	206.6 (33.4)	1.8 (0.28)	209.06 (34.9)	1.79 (0.29)
Hooker	29	186.3 (34.7)	1.76 (0.33)	181.8 (35.3)	1.7 (0.33)	190.71 (35.2)	1.8 (0.33)
Tighthead prop	32	210.0 (27.8)	1.71 (0.23)	211.5 (30.4)	1.7 (0.24)	208.46 (29.5)	1.7 (0.25)
Lock	39	199.7 (26.0)	1.69 (0.21)	197 (29)	1.7 (0.24)	202.84 (28.2)	1.72 (0.23)
Flanker	45	194.3 (25.7)	1.81 (0.22)	192.8 (28)	1.8 (0.24)	195.86 (26.8)	1.82 (0.22)
No. 8	19	203.9 (42.9)	1.77 (0.36)	198.2 (44.2)	1.7 (0.38)	210.58 (44.3)	1.82 (0.36)
Backs	153	184.2 (26.8) ^a	2.00 (0.28)	182.7 (27.7)	2 (0.29)	185.98 (29.1)	2.01 (0.3)
Scrum Half	31	176.9 (28.2)	2.09 (0.35)	176.5 (29.1)	2.1 (0.35)	177.29 (29.1)	2.1 (0.38)
Fly Half	20	170.0 (22.9)	1.93 (0.26)	167.4 (24.2)	1.9 (0.27)	172.69 (22.4)	1.96 (0.25)
Centre	41	191.4 (23.9)	1.95 (0.21)	188.6 (22)	1.9 (0.2)	194.29 (27.8)	1.98 (0.24)
Wing	38	191.7 (29.8)	2.02 (0.29)	192.4 (31.2)	2 (0.31)	192 (31.4)	2 (0.3)
Full back	21	180.9 (20.7)	1.97 (0.25)	177.7 (24.7)	1.9 (0.29)	184.21 (27.1)	2.01 (0.3)

^a Identifies the difference between the Backs and Forwards exceeds the MDC for the device.

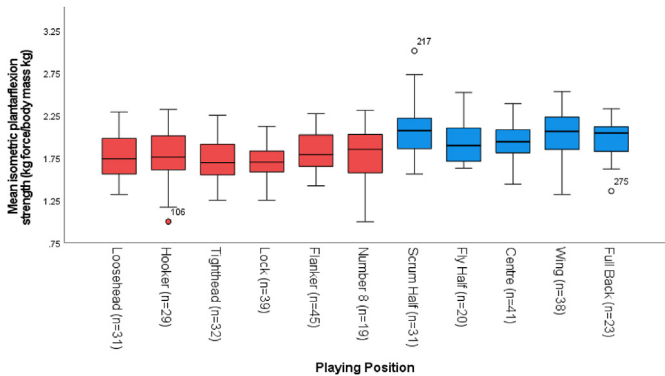


Fig. 2. Box plot of normalised isometric plantarflexion strength for players by playing position.

significantly more distance at high speed than tight forwards (Front and second row) (Lindsay et al., 2015). However, due to the intermittent nature of rugby union matchplay, average impact and running values fail to accurately illustrate the work done during the most intense periods of play for players in all positions. A key aim of athletic preparation is to condition players to be able to cope with these ‘worst case scenario’ periods and consequently work has been done attempting to quantify the demands of these highest intensity periods of play using rolling averages over shorter periods of time (Peek et al., 2021). When viewed through this lens, the

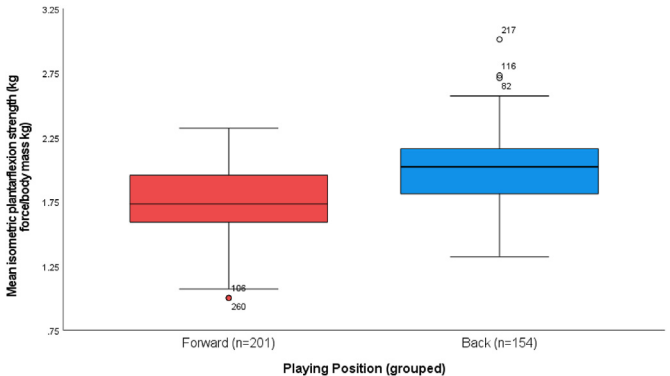


Fig. 3. Box plot of normalised isometric plantarflexion strength for players by playing group.

Table 4
Summary of average normalised combined limb strength by age group.

Age group (years)	Number	Normalised combined strength
18–22	154	1.88 (0.31)
23–28	118	1.84 (0.28)
29+	76	1.86 (0.28)

differences in positional demand are far less clear and

Table 5

Difference in average normalised combined limb strength by age group. The adjusted p-value is a result of post-hoc analysis using Tukey's method.

Age group (years)	Difference in mean normalised combined strength	95% Confidence Interval	Adjusted P-value
18-22 vs 23-28	0.038	(-0.046, 0.122)	0.537
18-22 vs 29+	0.014	(-0.083, 0.111)	0.935
23-28 vs 29+	-0.024	(-0.078, 0.024)	0.845

consequently, players of all positions may experience similar maximal impact and running intensities at some point in a given match (Peek et al., 2021). This is of clear relevance to practitioners in the field when considering athlete conditioning and injury risk mitigation strategies. The data from this study is presented by playing position to ensure it is of maximal practical utility to practitioners and athletes in the field.

Additional data analysed from the last four seasons of injuries in the English Premiership reveals that those players over 29 years of age sustain significantly higher rates of calf injury than those younger (Kemp et al., 2019). It is also thought that the calf strength of runners reduces as they age (Devita et al., 2016). We did not see a reduction in isometric calf strength in those players in our sample older than 29 years of age in relation to those younger (see Tables 4 and 5). It should be acknowledged that the strength scores within our dataset were taken in pre or very early season phase. It is not known whether strength scores change over the course of a competitive season with potentially resultant injury risks. This relationship has been shown in adductor muscle strength performance, which was seen to reduce in the weeks preceding the onset of injury in a group of Australian Football players (Crow et al., 2009). It seems possible that something similar may occur in the calf complex and future studies should build in more frequent testing. This would further inform practitioners of the likely increased risk to player groups during the course of a competitive season.

4.1. Study limitations

A limitation of the study was that players with previous history of calf or Achilles injury were not excluded from the sample. There is potential for those individuals to distort the strength scores of those with no history of injury, however previous work has identified limited differences post rehabilitation on measures of calf strength (O'Neill et al., 2019; Hébert-Losier et al., 2022). We chose to include all those deemed fit to train on the given testing day in order to obtain the most comprehensive picture of competing professional rugby athletes in England at this time. Future work should examine the impact of previous relevant injury history on isometric plantarflexion strength to provide context to clinicians in the field.

The study design pragmatically completed the strength testing alongside the normal activities of a training day at the clubs involved. Whilst all clubs completed testing on one of the first two training days of the week, the point in the day (and consequently the volume and type of physical training completed prior to testing) will have varied for subjects both within and between clubs. This has the potential to affect the results given that training activities were not standardised across all participants prior to testing. Again, future work into the effect of training related fatigue on isometric plantarflexion strength scores would provide valuable context to this area.

The strength data can only be viewed as a snapshot at the point of testing, given that no repeat testing within a training week, month or season was conducted, we cannot therefore assess how these isometric strength profiles change over periods of time.

Again, this would give useful context to those in the field and should be examined through future work.

It would also be valuable to replicate this research in a female professional rugby playing cohort. This would inform clinicians regarding the normative plantarflexion strength values of a female athlete population.

Some consideration needs to be given to the population sizes for playing positions as whilst most of our groups were large, we only sampled 19 number 8's.

4.2. Clinical implications

Plantarflexion strength is a key contributor to athletic performance in multidirectional running based sports such as rugby union (Hamner & Delp, 2013; Maniar et al., 2022; Marshall et al., 2014). It is also thought to be an important intrinsic risk factor in calf and Achilles injuries (Green et al., 2022; Mahieu et al., 2006). The data from this study can act as normative values of isometric calf strength in professional rugby union. This information is of significant value - informing position-specific athletic development programming, rehabilitation and Return to Play decisions across Premiership rugby union in England and likely internationally.

Normative values are particularly important as our previous work has identified that athletes with unilateral injuries have bilateral weakness when compared to normative data (O'Neill et al., 2019) and this is also supported by prospective data on injury development (Mahieu et al., 2006).

Clinically the isometric testing completed with our study offers a quick, valid and reliable tool that is relatively cheap in comparison to isokinetic testing however it does not give the full information about force curves and the ability of the tendon to contribute to energy recoil (Con:Ecc testing on isokinetics). We would therefore recommend with challenging rehabilitation cases more thorough testing procedures be considered.

5. Conclusion

This study is the first to publish isometric plantarflexion scores for professional rugby players and does so with position specific values from a relatively large cohort of players. The cohort comprised players from a broad range of different club environments across the English Premiership. Both of these factors improve the practical and external validity and generalisability of the data and thus the utility of it for fellow practitioners in the field.

This data provides normative values for professional rugby union players which may be applicable in performance profiling contexts, and may also provide more robust position-specific strength targets for players returning to play following injury than contralateral limb strength values. Practitioners may want to target soleus strength development due to increased rates of injury and potentially focus on monitoring the strength of this group more regularly over the course of a season.

Contributorship

MLe and SO'N both contributed equally to the design of the

work. SO'N controlled the data acquisition with substantial contribution from MLa, LT, BO'L, EP, DH, BS and AB. ML drafted the manuscript. FM contributed significantly to statistical analyses. SO'N, MLa, LT, BO'L, EP, DH, BS and AB critically reviewed the drafted manuscript. All authors gave final approval and agreed to be accountable for all aspects of the work. MLe is the guarantor of the study.

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Ethical approval information

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Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ptsp.2023.03.007>.

References

- Albracht, K., Arampatzis, A., & Baltzopoulos, V. (2008). Assessment of muscle volume and physiological cross-sectional area of the human triceps surae muscle in vivo. *Journal of Biomechanics*, 41, 2211–2218. <https://doi.org/10.1016/j.jbiomech.2008.04.020>
- Baxter, J., & Piazza, S. (2014). Plantar flexor moment arm and muscle volume predict torque-generating capacity in young men. *Journal of Applied Physiology*, 116, 538–544. <https://doi.org/10.1152/japplphysiol.01140.2013>
- Bridgeman, L. A., & Gill, N. D. (2021). The use of global positioning and accelerometer systems in age-grade and senior rugby union: A systematic review. *Sports Medicine - Open*, 7, 15. <https://doi.org/10.1186/s40798-021-00305-x> [published Online First: Feb 22].
- Brooks, J. H. M., Fuller, C. W., & Kemp, S. P. T. (2005). Epidemiology of injuries in English professional rugby union: Part 1 match injuries. *British Journal of Sports Medicine*, 39. <https://doi.org/10.1136/bjsm.2005.018135>
- 765–64 Brooks, J. H. M., & Kemp, S. P. T. (2011). Injury-prevention priorities according to playing position in professional rugby union players. *British Journal of Sports Medicine*, 45. <https://doi.org/10.1136/bjsm.2009.066985> [published Online First: Aug].
- Crow, J. F., Pearce, A. J., Veale, J. P., et al. (2009). Hip adductor muscle strength is reduced preceding and during the onset of groin pain in elite junior Australian Football players. *Journal of Science and Medicine in Sport*, 13, 202–204. <https://doi.org/10.1016/j.jsams.2009.03.007>
- Cuschieri, S. (2019). The STROBE guidelines. *Saudi Journal of Anaesthesia*, 13, 31–34. https://doi.org/10.4103/sja.SJA_543_18 [published Online First: Apr 01].
- Devita, P., Fellin Rebecca, E., Seah Joseph, P., et al. (2016). The relationships between age and running biomechanics. *Medicine & Science in Sports & Exercise*, 48, 98–106. <https://doi.org/10.1249/MSS.0000000000000744> [published Online First: Jan].
- Dorn, T. W., Schache, A. G., & Pandy, M. G. (2012). Muscular strategy shift in human running: Dependence of running speed on hip and ankle muscle performance. *Journal of Experimental Biology*, 215, 1944–1956. <https://doi.org/10.1242/jeb.064527> [published Online First: Jun 01].
- Ekstrand, J., Häggglund, M., & Waldén, M. (2011). Epidemiology of muscle injuries in professional football (soccer). *The American Journal of Sports Medicine*, 39. <https://doi.org/10.1177/0363546510395879> [published Online First: -02-18].
- Ferris, R. M., Hawkins, D. A., & Hawkins, D. (2020). Gastrocnemius and soleus muscle contributions to ankle plantar flexion torque as a function of ankle and knee angle. *Sports Injuries and Medicine*. <https://doi.org/10.29011/2576-5956.100063>
- Freckleton, G., Cook, J., & Pizzari, T. (2014). The predictive validity of a single leg bridge test for hamstring injuries in Australian Rules Football players. *British Journal of Sports Medicine*, 48, 713–717. <https://doi.org/10.1136/bjsports-2013-092356> [published Online First: Apr].
- Green, B., Lin, M., McClelland, J. A., et al. (2020b). Return to play and recurrence after calf muscle strain injuries in elite Australian Football players. *The American Journal of Sports Medicine*, 48, 3306–3315. <https://doi.org/10.1177/0363546520959327> [published Online First: Nov].
- Green, B., Lin, M., Schache, A. G., et al. (2020a). Calf muscle strain injuries in elite Australian football players: A descriptive epidemiological evaluation. *Scandinavian Journal of Medicine & Science in Sports*, 30, 174–184. <https://doi.org/10.1111/sms.13552> [published Online First: Jan].
- Green, B., McClelland, J. A., Semciw, A. I., et al. (2022). The assessment, management and prevention of calf muscle strain injuries: A qualitative study of the practices and perspectives of 20 expert sports clinicians. *Sports Medicine - Open*, 8, 10. <https://doi.org/10.1186/s40798-021-00364-0> [published Online First: Jan 15].
- Green, B., & Pizzari, T. (2017). Calf muscle strain injuries in sport: A systematic review of risk factors for injury. *British Journal of Sports Medicine*, 51, 1189–1194. <https://doi.org/10.1136/bjsports-2016-097177> [published Online First: Aug].
- Häggglund, M., Waldén, M., & Ekstrand, J. (2013). Risk factors for lower extremity muscle injury in professional soccer. *The American Journal of Sports Medicine*, 41, 327–335. <https://doi.org/10.1177/0363546512470634> [published Online First: Feb].
- Hamner, S. R., & Delp, S. L. (2013). Muscle contributions to fore-aft and vertical body mass center accelerations over a range of running speeds. *Journal of Biomechanics*, 46, 780–787. <https://doi.org/10.1016/j.jbiomech.2012.11.024>
- Hébert-Losier, K., Newsham-West, R. J., Schneiders, A. G., et al. (2009). Raising the standards of the calf-raise test: A systematic review. *Journal of Science and Medicine in Sport*, 12, 594–602. <https://doi.org/10.1016/j.jsams.2008.12.628> [published Online First: Nov 01].
- Hébert-Losier, K., Ngawhika, T. M., Gill, N., et al. (2022). Validity, reliability, and normative data on calf muscle function in rugby union players from the Calf Raise application. *Sports Biomechanics*. <https://doi.org/10.1080/14763141.2022.2118158> [published Online First: -09-19].
- Houghton, L., Dawson, B., & Rubenson, J. (2013). Achilles tendon mechanical properties after both prolonged continuous running and prolonged intermittent shuttle running in cricket batting. *Journal of Applied Biomechanics*, 29, 453–462. <https://doi.org/10.1123/jab.29.4.453> [published Online First: Aug].
- Kawakami, Y., Muraoka, T., Ito, S., et al. (2002). In vivo muscle fibre behaviour during counter-movement exercise in humans reveals a significant role for tendon elasticity. *Journal of Physiology*, 540, 635–646. <https://doi.org/10.1113/jphysiol.2001.013459> [published Online First: Apr 15].
- Kemp S, Starling L, Anstiss T, et al. England Professional Rugby Injury Surveillance Project Report 2019–20. <https://www.englandrugby.com/dxdam/ab/ab1ea449-5915-4c5c-ab27-9f90ed076bd8/PRISP%20report%2019-20%20Final.pdf>.
- Kemp S, West S, Brooks J, et al. England professional rugby injury surveillance project 2017–2018 Season Report. https://www.englandrugby.com/dxdam/96/960006d9-269d-4250-a15f-d9e62f8bfe70/PRISP_1718.pdf.
- Kemp S, West S, Brooks J, et al. England professional rugby injury surveillance project 2018–2019. <https://www.englandrugby.com/dxdam/89/8994c1ba-17df-47a3-a4c6-8878ed6728d0/England%20PRISP%2018.19.pdf>.
- Kulmala, J., Korhonen, M. T., Ruggiero, L., et al. (2016). Walking and running require greater effort from the ankle than the knee extensor muscles. *Medicine & Science in Sports & Exercise*, 48, 2181–2189. <https://doi.org/10.1249/MSS.0000000000001020> [published Online First: Nov].
- Laufer, B., Lichtwark, G. A., & Cresswell, A. G. (2014). Reciprocal activation of gastrocnemius and soleus motor units is associated with fascicle length change during knee flexion. *Physiological Reports*, 2, Article e12044. <https://doi.org/10.14814/phy2.12044> [published Online First: Jun].
- Lindsay, A., Draper, N., Lewis, J., et al. (2015). Positional demands of professional rugby. *European Journal of Sport Science*, 15. <https://doi.org/10.1080/17461391.2015.1025858> [published Online First: -04-01].
- Mahieu, N. N., Witvrouw, E., Stevens, V., et al. (2006). Intrinsic risk factors for the development of achilles tendon overuse injury. *The American Journal of Sports Medicine*, 34, 226–235. <https://doi.org/10.1177/0363546505279918> [published Online First: Feb].
- Maniar, N., Cole, M. H., Bryant, A. L., et al. (2022). Muscle force contributions to anterior cruciate ligament loading. *Sports Medicine*. <https://doi.org/10.1007/s40279-022-01674-3>
- Mansournia, M. A., Collins, G. S., Nielsen, R. O., et al. (2021). A Checklist for statistical assessment of medical Papers (the CHAMP statement): Explanation and elaboration. *British Journal of Sports Medicine*, 55. <https://doi.org/10.1136/bjsports-2020-103652>
- Marshall, B. M., Franklyn-Miller, A. D., King, E. A., et al. (2014). Biomechanical factors associated with time to complete a change of direction cutting maneuver. *The Journal of Strength & Conditioning Research*, 28, 2845–2851. <https://doi.org/10.1519/JSC.0000000000000463> [published Online First: Oct].
- O'Neill, S., Barry, S., & Watson, P. (2019). Plantarflexor strength and endurance deficits associated with mid-portion Achilles tendinopathy: The role of soleus. *Physical Therapy in Sport*, 37, 69–76. <https://doi.org/10.1016/j.ptsp.2019.03.002> [published Online First: May].
- O'Neill, S., Weeks, A., Nørgaard, J., & Jørgensen, M. (2023). *Validity and intrarater reliability of a novel device for assessing plantar flexor strength*. PLOS ONE - In Press.

- O'Neill, S., Radia, J., Bird, K., et al. (2019). Acute sensory and motor response to 45-s heavy isometric holds for the plantar flexors in patients with Achilles tendinopathy. *Knee Surgery, Sports Traumatology, Arthroscopy*, 27, 2765–2773. <https://doi.org/10.1007/s00167-018-5050-z> [published Online First: Sep 01,].
- Pandy, M. G., Lai, A. K. M., Schache, A. G., et al. (2021). How muscles maximize performance in accelerated sprinting. *Scandinavian Journal of Medicine & Science in Sports*, 31, 1882–1896.
- Peek, R. J., Middleton, K. J., Gastin, P. B., et al. (2021). Position specific peak impact and running demands of professional rugby union players during game play. *International Journal of Sports Science & Coaching*, 16, 1162–1168. <https://doi.org/10.1177/17479541211004065> ([published Online First: Oct]).
- Waterworth, G., Wein, S., Gorelik, A., et al. (2017). MRI assessment of calf injuries in Australian football League players: Findings that influence return to play. *Skeletal Radiology*, 46, 343–350. <https://doi.org/10.1007/s00256-016-2564-7> [published Online First: Jan 16,].
- Wellman, A. D., Coad, S. C., Goulet, G. C., et al. (2016). Quantification of competitive game demands of NCAA Division I college football players using global positioning systems. *The Journal of Strength & Conditioning Research*, 30, 11–19. <https://doi.org/10.1519/JSC.0000000000001206> ([published Online First: Jan]).
- Werner, B. C., Belkin, N. S., Kennelly, S., et al. (2017). Acute gastrocnemius-soleus complex injuries in National Football League athletes. *Orthopaedic Journal of Sports Medicine*, 5, <https://doi.org/10.1177/2325967116680344> [published Online First: -01-01].
- Willer, J., Allen, S. J., Burden, R. J., et al. (2021). Neuromechanics of middle-distance running fatigue: A key role of the plantarflexors? *Medicine & Science in Sports & Exercise*, 53, 2119–2130. <https://doi.org/10.1249/MSS.0000000000002695> ([published Online First: Oct]).